

Concerns and Redefining Load Modeling of Large Data Centers in Future Power Systems—A Review

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Abstract—The exponential rise of generative artificial intelligence is driving an unprecedented surge in hyperscale data center capacity, creating novel stability challenges for future power systems. Traditional load modeling approaches, largely based on induction motor and static load counterparts, are inadequate for capturing the complex, control-driven dynamics of modern rectifier-interfaced data center facilities. This paper discusses data center behavior from a system perspective, identifying critical risks such as transient instability caused by weak fault ride-through capabilities and oscillatory instability driven by periodic artificial intelligence workload patterns. We critique existing static and composite load models and suggest the requirement of a decision framework for selecting appropriate modeling domains, based on the characteristics of the phenomena under study. Finally, the paper outlines urgent industry needs, emphasizing standardized blackbox modeling to overcome intellectual property barriers and the necessity of high-resolution field data for model validation to ensure power system integrity.

Index Terms—Data centers, Load modeling, Oscillations, Power system dynamics, Power system stability

I. INTRODUCTION

The generative artificial intelligence (AI) boom has inevitably led to an upcoming proliferation of large scale data centers (DCs) both nationally and globally. Enabling technologies like big data analytics, high performance cloud computing have caused this growing trend in demand for large DCs. Unprecedented load growth driven by generative AI has shifted the focus from existing enterprise-type DCs to hyperscale facilities. Such a facility could easily consume 100 MW, if not multiples of hundreds MW of electric power on a regular basis [1]. Hence, the modern DCs are becoming key players in future power systems.

Currently, the USA is the global leader in terms of the number of the active DCs and the amount of data traffic handled by them. Northern Virginia, Atlanta (Georgia), Dallas (Texas), Silicon Valley (California) are some of the notable hotspots for construction of new DCs in the US [2]. According to a recent study by western electricity coordinating council (WECC), DCs make up for 78% of the new large load interconnection requests queue in the region [3]. The total load demand from these upcoming DCs is over 90% of the current system peak demand. Similarly, 73% of the 2025 large load queue in the ERCOT region comprise of DCs [4]. As shown in Fig. 1, the McKinsey & Company predicts that the global DC capacity demand will rise at an compound annual growth rate

(CAGR) of 22% from 2023 to 2030, where the generative AI workloads will be the key driver with a CAGR of 39% alone [5].

With these trends, comes a host of risks and challenges to the power system stability and reliability. The grid has already started experiencing some signs of stress that a set of large DC loads can cause. For instance, on July 10, 2024, a fault on a 230 kV transmission line led to a subsequent sudden disconnection of 1500 MW of DC load from the grid [6] that nearly initiated a *byte blackout* in the Dominion Energy service area of PJM interconnection. This incident was flagged as an extreme outlier against typical load swings in the region [7]. From a recent whitepaper from North American Electric Reliability Corporation (NERC) [8], Figure 2 shows the frequency response of the eastern interconnection during this incident in a six-minute window. The frequency had risen above the upper deadband of +0.036 Hz and stayed above it for nearly a minute, which is a clear indication of the severity of the event in terms of stability of the bulk power system (BPS). Similar incidents related to large loads have been reported in the ERCOT region too [8].

These factors of rapid growth of large DC loads and their emerging impacts on power system stability have made it

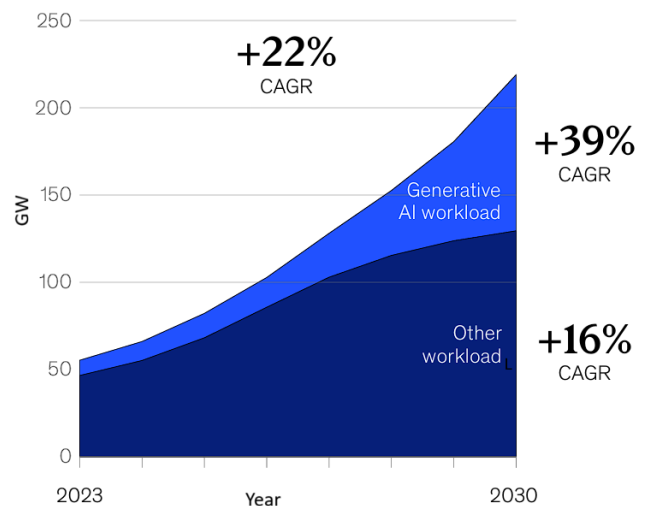


Fig. 1. Estimated global data center capacity demand [5]

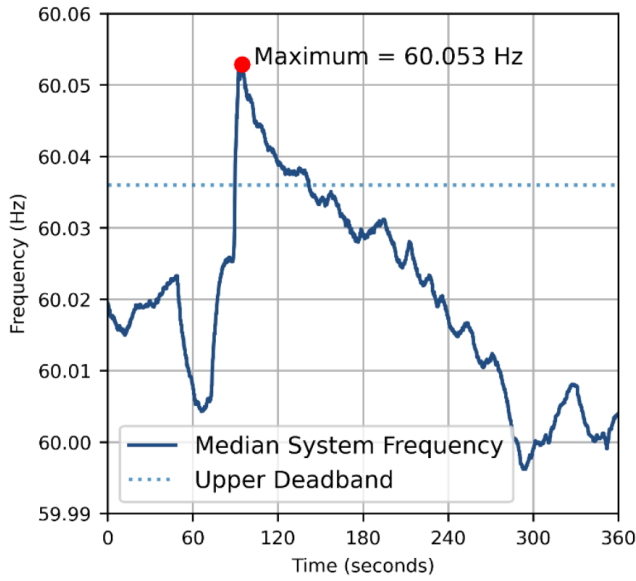


Fig. 2. Frequency plot for Eastern interconnection during the 1500 MW of primarily data center load transfer incident [8]

imperative to study their dynamic characteristics and behavior from a system perspective. We are reaching a point where the traditional load modeling approaches would not be sufficient to capture the dynamic phenomena associated with such large loads [9]. It is crucial to repurpose and enhance existing load models or develop new ones that can accurately represent the behavior of large DC loads under various operating conditions. This paper aims to provide insights into the requirement of new dynamic load modeling approaches for large DC loads in future power systems. It is noteworthy that this paper focuses on the system perspective of large DC loads rather than the efficiency or effectiveness of the internal DC operations.

The paper is organized as follows: Section II discusses the technical characteristics of a large DC load. Section III delves into the dynamic phenomena and system stability risks associated with large DC loads. Section IV explores load modeling considerations for large DC loads. Section V outlines gaps and future needs pertaining to this area. Finally, Section VI concludes the paper.

II. TECHNICAL CHARACTERISTICS OF A LARGE DATA CENTER LOAD

A. Electrical Infrastructure of a Data Center

A DC consists of several key components that work together to ensure efficient data processing, storage, and management. These components include servers, networking equipment, storage systems, cooling systems, power supply units, and more. Figure 3 depicts a simplified configuration of a typical DC. Regardless of the type (conventional enterprise DCs or hyperscale DCs), the fundamental components remain similar, although the scale and complexity may vary significantly. A DC primarily consists of the following components that either consume or supply electrical power [10]:

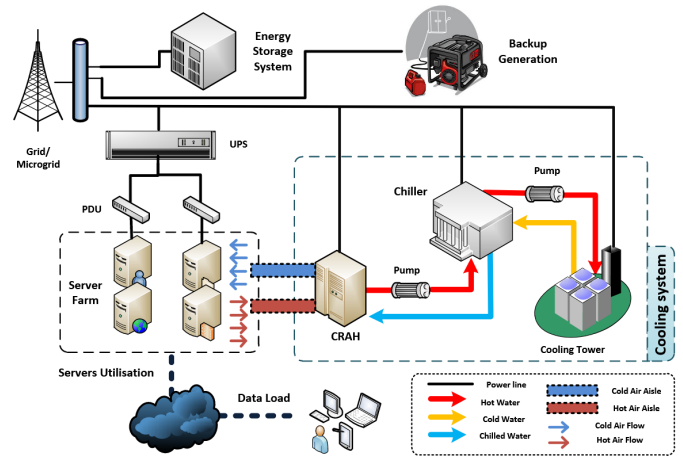


Fig. 3. Components in a typical data center [10]

- 1) *Server farm*: The core of a DC, consisting of racks of servers that handle ICT service.
- 2) *Power supply equipment*: This includes power conditioning devices like uninterruptible power supply (UPS) systems, power distribution units (PDUs). The primary source of power is typically the utility grid that is connected to the DC through a high voltage or medium voltage transformer. This utility connection is usually connected to the UPS system with a bypass switch to ensure continuous power supply to the DC in case of grid disturbances.
- 3) *Backup generation*: While UPS units are meant to provide the DC with power for short periods of time, backup generation from diesel generation may provide power for prolonged periods in case of utility grid failures.
- 4) *Energy storage*: To match with the high flexibility of the overall load demand from a DC, battery systems are usually used as further backup, that may be coupled with renewable energy sources.
- 5) *Cooling system*: It is common for a DC to have a water chiller plant-based cooling system accompanied by a computer room air handler (CRAH). CRAH ensures that the ambient temperature of the servers is within an acceptable range.

Among the electrical elements listed above, the *server farm*, and the *cooling (air conditioning) system* constitute the major portion of a large DC load. Depending on the scale of the DC, power consumption by the servers can be as high as 80% of the total load of the DC facility [11].

B. Utility Grid Interface Behavior

After the PDUs distribute ac power towards the server farm, power supply units (PSUs) convert ac power to dc power by means of rectifiers, and feed the power to servers [12]. However, some modern DCs may have dc PDUs. These rectifier-based electronic loads employ stringent power quality requirements such that the DC is set to switchover from the utility grid to local UPS-based power in case of grid

disturbances that cause voltage sags in the grid. This voltage sensitivity that enforces power quality requirements in the DC facility limits the fault ride-through (FRT) capability of the DC facility [11].

C. AI Workload Profiles and Consumption Patterns

The behavior of AI workload profiles is a key factor that distinguishes a hyperscale AI DC from a conventional DC. The magnitude of load fluctuations of a single large DC or a few large DCs could be multiple hundreds MW, if not well over 1000 MW, reflecting large and undesirable load swings on the BPS in the near future [3]. These AI workload patterns are governed by the fundamental nature of AI processes driven by graphical processing units (GPUs). This can be conceptually illustrated as shown in Fig. 4.

The AI workload patterns could even vary between the stages of AI computing [13]: *Training*, *Fine-tuning*, and *Inference*. This can be observed in Fig. 5 that shows these various AI computing patterns on a single GPU, where a large DC would have hundreds of thousands of such GPUs. AI workloads differ significantly from traditional cloud computing. Training workloads cause sustained high demand with periodic fluctuations, while inference workloads create *bursty*, stochastic power spikes.

III. DYNAMIC PHENOMENA AND SYSTEM STABILITY RISKS OF LARGE DATA CENTER LOADS

We can identify three facets of large DC loads that could create the problems in the power grid:

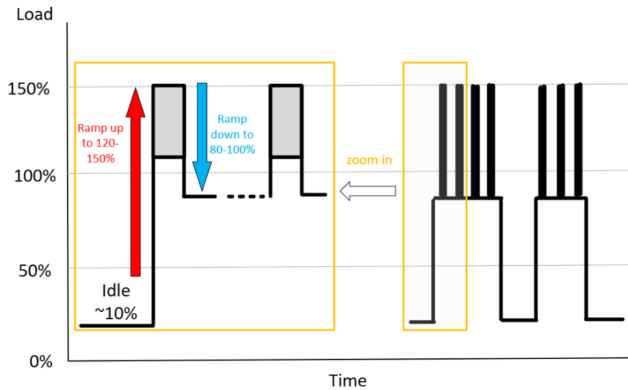


Fig. 4. Variability and ramping of AI demand profile [3]

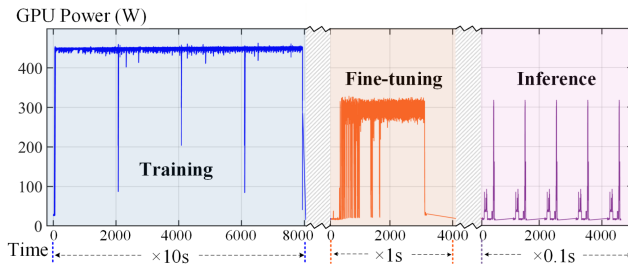


Fig. 5. Different AI computing load patterns in a single GPU [13]

- 1) **Inherent dynamics** of the DC that encompass large rectifier-based electronic load elements, and control and protection dynamics.
- 2) **FRT capability** of the DC that is governed by power quality enforcement or over-sensitive power conditioning equipment.
- 3) **Variability of AI workloads** that is governed by user activity patterns.

Some of the prominent stability risks that emerge from above factors are as follows,

A. Transient Stability

As briefly discussed in Section I, Eastern interconnection and ERCOT have experienced massive load swing incidents that is caused by weak FRT behavior of large DC loads. The integration of hyperscale DCs introduces distinct transient stability risks driven by the voltage sensitivity of these facilities. Unlike traditional industrial loads dominated by induction motors (IMs), which may stall but often remain connected during voltage sags, rectifier-interfaced DCs tend to switchover from the grid to local power supply. Figure 6 shows a positive-sequence, phasor domain simulation results of two generators losing synchronism and attaining a rotor angle instability due to a post-fault tripping of a nearby DC load [8].

B. Oscillatory Stability

The integration of hyperscale DCs introduces novel oscillatory stability risks that differ fundamentally from traditional load behaviors. These risks manifest through two primary mechanisms: *negative damping or resonance* caused by the control interactions of rectifier-based power electronic loads with the grid, and *forced oscillations* driven by the periodic nature of AI computational workloads [8]. We can categorize them as subsynchronous oscillations (SSOs).

1) *Resonance-based oscillations*: In 2017, low frequency resonance phenomenon of ~ 11 Hz was observed in a set of Meta DCs [12], which was clearly visible as an envelope

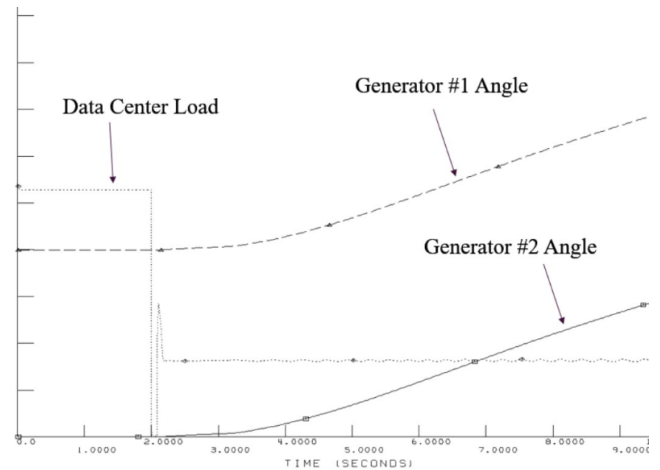


Fig. 6. Simulation of loss of synchronism of generators after post-fault tripping of nearby data center [8]

waveform in one of the representative measurement scopes as shown in Fig. 7, in both current and voltage waveforms. The study [14] takes this subsynchronous resonance concern one step further by investigating a ~ 14.7 – 14.8 Hz oscillations coming from a DC in Dominion Energy's power system. Furthermore, with the help of non-parametric spectral analysis on historical data of similar grid instances, using power spectral densities (PSDs), the study frames a *critical mode of the DC in the range of 10–11 Hz with significantly low damping* to be the root cause of the oscillations at ~ 14.7 – 14.8 Hz.

2) *Forced oscillations*: Unlike traditional stochastic loads, AI training and fine-tuning workloads exhibit highly structured, periodic power consumption profiles driven by synchronized Graphics Processing Unit (GPU) operations [8]. Using a modified WECC 179-bus system with multiple DCs having stochastic AI workload profiles, [15] demonstrates that DC-induced forced oscillations are not merely local; they can also manifest as inter-area oscillation patterns as shown in Fig. 8, where GENROU 9–12 are generators of the system.

C. Power Quality and Harmonics

Unlike linear loads, DCs rely heavily on non-linear components, specifically, rectifier-based PSUs for IT server racks and variable frequency drives (VFDs) for cooling systems, which draw non-sinusoidal currents and inject harmonic distortions into the grid. These raise power quality concerns originating from large DCs [1]. If not mitigated by active filtering,

these currents cause non-linear voltage drops across system impedances, resulting in voltage distortion that can propagate to neighboring industrial and residential customers [13]. In addition to low frequency resonance that was discussed previously, high frequency resonance phenomena have been observed in recent field observations in the 5–10 kHz range [1]. Regardless of the range of these low or high frequency resonances, they can easily lead to catastrophic instability when subjected to grid disturbances. Other power quality concerns like voltage flicker are also becoming more common with the penetration of new large DC loads [8].

IV. LOAD MODELING CONSIDERATIONS FOR LARGE DATA CENTER LOADS

The anticipated rapid penetration of hyperscale DC loads necessitates a fundamental shift in load modeling practices. While traditional modeling frameworks have sufficed for decades, the distinct electrical characteristics of DCs as highlighted by preceding sections—largely dominated by power electronics rather than rotating machines—require new methodologies to capture their interaction with the BPS [9].

Historically, power system modeling prioritized the detailed representation of generation resources, while loads were approximated using simplified static representations. Planners relied on constant impedance, current, and power (ZIP) models, assuming these algebraic representations were sufficient for bulk system assessments. This paradigm shifted following the August 10, 1996, WECC blackout, which revealed that static models failed to capture the delayed voltage recovery caused by the stalling of IMs [16]. In response, the industry developed the *composite load model* (CLM). This framework combines static load components with differential equation-based IM models, allowing planners to simulate the transient response of motor-dominated industrial and residential loads [17]. While the CLM represented a significant advancement for motor loads, it represents electronic loads merely as a static constant power component. This representation is increasingly inadequate for modern hyperscale DCs, where server racks fed by power electronic rectifiers constitute approximately 80–90% of the total load, with cooling motors comprising only the remaining minority share.

The core limitation of applying existing industrial load models to DCs lies in the source of their dynamic behavior. The transient response of traditional industrial facilities is governed by the physics of IMs, which is consistent across manufacturers and well-understood. In contrast, the response of a DC is governed by the proprietary control (and protection) logic of its PSUs and UPS systems. Because these devices are software-defined, their behavior during grid disturbances—such as FRT, current limiting, and voltage recovery—can vary vastly depending on the manufacturer and the specific programming of the active front-end rectifiers. Current static models fail to capture these complex control loops, often predicting unrealistically fast fault recovery or failing to simulate the disconnection of loads during voltage sags. Consequently, high-fidelity modeling for DCs requires

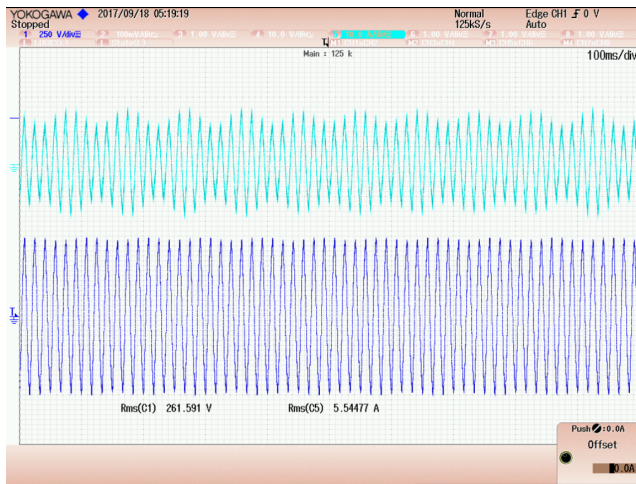


Fig. 7. Measured low frequency resonance in a data center in 2017 [12]

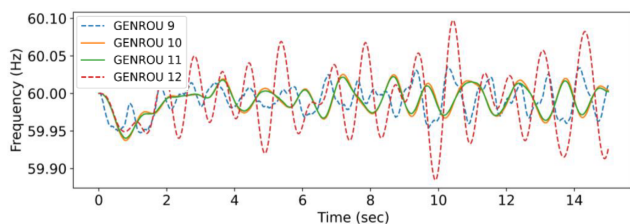


Fig. 8. Inter-area oscillation between generators, induced by data centers [15]

moving beyond *physics-based* assumptions to *control-based* representations [3], [8], [18].

The selection of the appropriate modeling domain depends on the bandwidth of the phenomena being studied. The choice may come in three different approaches: *phasor domain*, *electromagnetic transient (EMT)*, and *hybrid approach* [13]:

1) *Phasor domain modeling*: For BPS reliability studies focusing on electromechanical dynamics (0.1–10 Hz), aggregated phasor-domain models are generally sufficient. These models can represent the aggregate active and reactive power response of the facility to fundamental frequency voltage variations.

2) *EMT modeling*: For assessments involving fast control interactions, high-frequency resonance, low-frequency resonance (SSO), or weak grid interconnects, detailed EMT models are required. EMT simulation is particularly critical for evaluating protection systems and assess the capability of the load that it can ride-through unbalanced faults without tripping.

3) *Hybrid modeling*: To trade-off between computational burden and accuracy, transmission planners and operational planners may employ a hybrid approach using voltage-dip contours. In this methodology, detailed dynamic models (CLM or EMT) are applied only to loads located within a specific voltage-dip threshold (e.g., >70% dip) relative to a fault, while loads outside this contour remain represented by efficient static models [19].

In general, the paramount importance is capturing all critical dynamics within the particular power system assessment. EMT approach could become an overkill due to its demanding modeling effort and high computational burden for a large

system. More accurate EMT models derived for operational planning may be used to parametrize phasor domain models to capture the slower dynamics which are of concern for BPS stability studies. The article [18] suggests a conceptual *threshold* which would act as a metric to decide when detailed models are needed so that the required dynamics would be captured with confidence. Inspired by [18] and [19], Fig. 9 presents a simple decision tree indicating when to use each of above approaches.

Modeling every individual server rack within a hyperscale DC facility is computationally prohibitive. Effective modeling relies on aggregating components based on the hierarchy of the facility's power distribution system. For instance, thousands of single-phase PSUs and three-phase UPS units can be aggregated into a single equivalent converter model that captures the facility's total electronic load dynamics [18]. The CLM design philosophy can be adapted to aggregate either *load components within a DC facility for plant-level studies*, or *multiple coherent DCs for BPS-level assessments*. To address the AI workload concerns of large DC load modeling, investigating the fusion of continuous and event-driven dynamics via a stochastic process-based approach [20] would be transformative.

V. FUTURE NEEDS

The integration of hyperscale DCs represents a paradigm shift for grid planning (both long term and operational planning), yet the industry lacks the standardized frameworks required to model these loads with the same rigor applied to generation resources. To ensure reliability, the following future

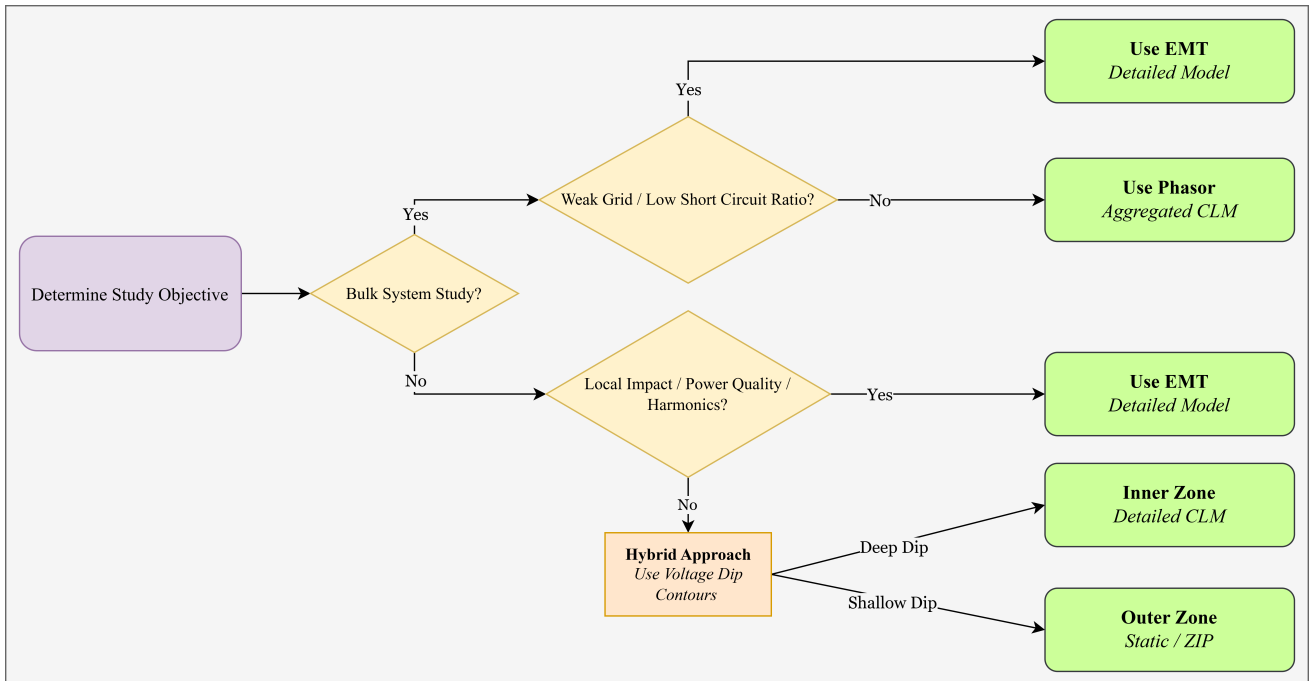


Fig. 9. Decision tree to select between phasor, EMT, and hybrid load modeling approaches

needs regarding data transparency, interconnection processes, and model validation must be addressed.

A primary barrier to accurate large DC load modeling is the proprietary nature of DC equipment. Unlike standard IMs, the dynamic response of a DC is governed by the proprietary control logic of PSUs and UPS units, which manufacturers are often reluctant to disclose due to intellectual property (IP) concerns. To overcome IP barriers, future standards should adopt frameworks similar to those used by the Australian Energy Market Operator (AEMO) [21]. AEMO guidelines explicitly allow and outline procedures for *blackbox-ing*, *compiling*, or *encrypting* EMT models. This approach allows manufacturers to protect their control algorithms while providing power system planners with a functional model that accurately represents the device's interaction with the grid. In the absence of detailed models during the early screening phases, transmission operators can utilize targeted questionnaires to gather critical technical data.

The pace of DC deployment significantly outstrips the timeline for transmission planning and model development. While DCs can be built and energized in 12 to 24 months, transmission infrastructure often requires 5 to 10 years to complete. To manage this disparity, future planning frameworks may need to adopt *connect and manage* approaches or expedited study timelines.

There is a critical lack of event-based measurement data for large DC loads, which hinders the ability to extrapolate the load dynamics and validate simulation models. Without high-resolution field data from actual grid disturbances, engineers cannot verify if the dynamic behavior predicted by models matches reality.

VI. CONCLUSION

The proliferation of hyperscale data centers—spurred by the estimated 39% compound annual growth rate of global generative AI workloads—has transformed these facilities into pivotal participants in future power systems. This paper highlights how traditional load modeling is inadequate for capturing the complex, control-driven dynamics inherent in large rectifier-interfaced data center loads. As evidenced by recent grid disturbances, high voltage sensitivity and weak fault ride-through capabilities present substantial transient stability risks to bulk power systems. Moreover, the inherent nature of GPU-driven AI computing introduces novel oscillatory challenges, specifically forced oscillations and resonance phenomena. Addressing these vulnerabilities necessitates a fundamental shift toward high-fidelity, control and protection-based dynamic load modeling frameworks that leverage phasor, electromagnetic transient, or hybrid domains. Ultimately, maintaining long-term grid security depends on overcoming intellectual property barriers through standardized blackbox modeling, and establishing rigorous validation protocols using high-resolution event data. Standardizing these methodologies is essential for ensuring power system integrity amidst the rapid, technology-driven evolution of global demand.

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